

TOK Exhibition

#23: How important are material tools in the production or acquisition of knowledge

(Word Count: 949)

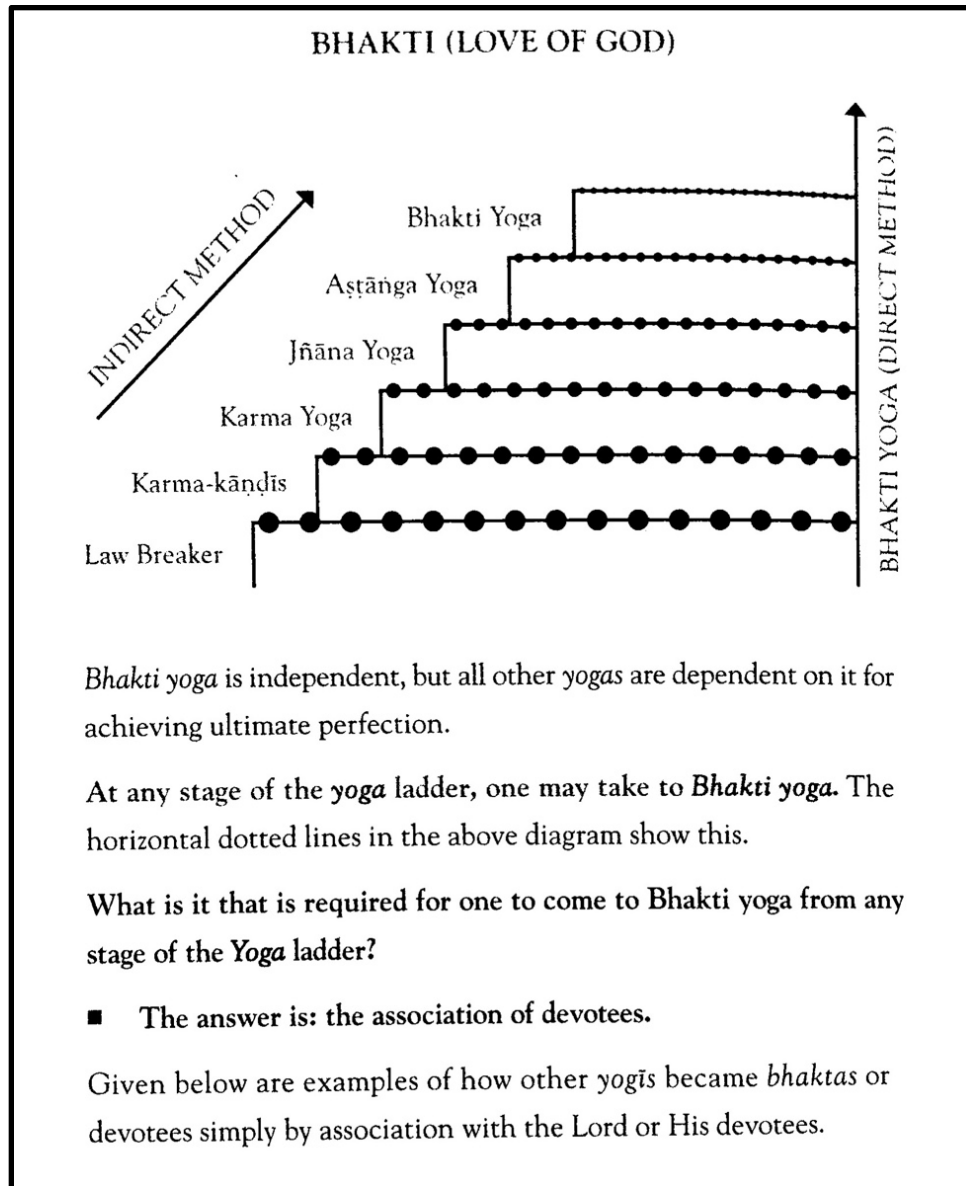
```
56 from sklearn.ensemble import RandomForestClassifier
57
58 # Initialize variables to track the best model
59 best_acc = 0
60 train_scores, test_scores = [], []
61
62 # Loop over the folds
63 for i, (train_index, test_index) in enumerate(kf.split(X_train_full)):
64     # Split the training data into training and testing sets for the fold
65     X_train, X_test = X_train_full[train_index], X_train_full[test_index]
66     y_train, y_test = y_train_full[train_index], y_train_full[test_index]
67
68     # Train the random forest classifier
69     rf_clf = RandomForestClassifier(n_estimators=11, max_depth=10, random_state=42)
70     rf_clf.fit(X_train, y_train.flatten())
71
72     # Predict the labels for the test set
73     y_pred = rf_clf.predict(X_test)
74
75     # Calculate accuracy for training data
76     y_train_pred = rf_clf.predict(X_train)
77     train_acc = accuracy_score(y_train, y_train_pred)
78
79     # Calculate accuracy for testing data
80     test_acc = accuracy_score(y_test, y_pred)
81
82     # Print accuracies
83     print(f"Fold {i}, Training Accuracy: {train_acc:.4f}, Testing Accuracy: {test_acc:.4f}")
84
85     # Append accuracies to track them across folds
86     train_scores.append(train_acc)
87     test_scores.append(test_acc)
88
89     # Store accuracy for the best model
90     if test_acc > best_acc:
91         best_acc = test_acc
92         best_model = rf_clf # Save the best model
93         best_i = i
94
95 # Print the best fold and its accuracy
96 print(f"Best fold: {best_i}, Best Cross-Validation Accuracy: {best_acc:.4f}")
```

Object 1: Screenshot Taken by Author of Machine Learning Algorithm used to Predict Progression of Parkinson's Disease

My first object is a machine learning program that I coded to try to predict the progression rate of Parkinson's Disease (PD). This model identified patterns in a large dataset with a list of lots of patient data, to eventually highlight subtle patterns such as dyskinesias that aren't easily detectable through traditional analysis. This helped medical experts understand the best way to treat PD.

This object links to the prompt because it suggests that technological tools have great importance in producing medical knowledge, particularly in identifying patterns that would have been previously undecipherable by humans. Machine learning (ML) is a subfield of AI that uses algorithms to make predictions based on data. Most datasets consist of ‘hundreds of thousands of datapoints’ which would be incomprehensible to humans, but ML helps uncover insights and patterns in these large datasets (Lobo, 2008). Our ML model produced probabilistic, evidence-based, and practically applicable knowledge, and particularly uncovered that dyskinesias (involuntary twitching of muscles) was a key indicator in assessing PD symptoms. Furthermore, line 86 in the object tests the accuracy of the ML model and ensures the most accurate model was used to assist patients. Evidence supports that a ML model provides a higher degree of certainty than its human counterparts (Warmerdam, n.d.). Thus, not only did this material tool uncover new markers to help patients, but because the medical knowledge produced provides a high degree of certainty, it is more likely to be trusted by a general community of knowers and would thus have greater implications to society.

Interestingly, this object helps consider how material tools can be misused. If not verified before its implementation, ML models can hallucinate and can produce incorrect insights that would mislead medical experts. Thus, to ensure that inaccurate knowledge doesn’t disseminate within communities of knowers, the results need to be cross validated and externally validated to reduce the risk of misinformation.



Object 2: Picture Taken by Author of Text and Diagram from a Section of a Page in the Book Called 'Basics of Bhagavad Gita' Printed in 2018

My second object is a section from a page in the *Bhagavad Gita*, which is a sacred Hindu scripture that is read by anyone studying Hindu philosophy and was passed down for centuries through oral tradition. Specifically, this section discusses the abstract concept of *Bhakti* (the process of achieving enlightenment) and includes text and a diagram to illustrate this concept.

This object links to the prompt because it suggests that religious tools can play a crucial role in acquiring spiritual knowledge. Religious texts are material tools that allow the preservation and dissemination of spiritual knowledge across generations. Particularly, the process in achieving Bhakti is complicated, involving numerous abstract concepts like Karma and Jnana Yoga which are difficult to understand. Oral dissemination of abstract concepts is less understandable, and a physical tool of a book can make it easier for a community of knowers to interpret the knowledge. Also, by using a physical book the knowledge is preserved in this current state and prevents any personal modifications that would happen in oral dissemination. Moreover, this object utilizes diagrams. The simple stepwise ladder schematic, demonstrating the direct method to enlightenment through Bhakti yoga, turns a complicated process into a simple diagram. Recent evidence supports this phenomenon as it was observed that written tools of abstract concepts preserve old knowledge (Qizi & Jumaeva, 2024). Thus, the physical book with a diagram allows communities of knowers to easily acquire and engage with spiritual knowledge and thus this knowledge remains shared rather than restricted to experts.

Furthermore, this object helps us explore the limitation on relying on experts to help us acquire knowledge. This object is written in Sanskrit originally which prevented other communities from acquiring knowledge due to the language barrier. The reliance on experts to translate this knowledge in other languages might create barriers to understanding as experts can risk the selective interpretation and presentation of knowledge.



Object 3: Image Taken by Author of the First 4 lines Of a Guitar Piece Titled *Recuerdos de la Alhambra* by Tárrega, Published in 1997

My third object is an image of the first 3 lines of a classical guitar piece by Tárrega. This is a technically challenging piece due to the unique technique required to play repeated 32nd notes called tremolo. However, after practicing this piece I started to implement my own creative ideas and express emotion in this piece.

This object links to the prompt because it suggests that artistic tools are secondary to personal experience in acquiring practical artistic knowledge as proficiency grows. Artistic knowledge refers to the application of technical skill that allows for emotional expression. The piece uses repeated tremolo throughout, and when learning the piece with little artistic experience, material tools such as the sheet music facilitate artistic production providing structure in the specific notes to play the piece musically correct. However, the rigidity of following the specific notes on the page restricts the interpretation of such knowledge. After I practiced the tremolo, I realized this repetition became monotonous for audiences. To make this piece more engaging and to express

emotion, I began changing the notes and the tempo using my personal experience which weren't indicated on the sheet music. Thus, representing how as the knower gains proficiency, their personal experience begins to play a greater role in shaping the artistic knowledge produced. Evidence supports this, as musician Charlie Parker stated how musicians utilize personal experience to create emotion (*A Quote by Charlie Parker*, n.d.). Ultimately, at a certain expertise, most pieces require expression and emotion, which can't be derived from artistic tools.

Additionally, this object considers the limitations in solely relying on personal experience to acquire knowledge. Artistic knowledge is often embodied and performative, involving muscle memory, expression and physical practice, which requires expert influence to guide the interaction between the tool technique and emotion. When learning this piece, my teacher provided specific finger placements which highlighted how expert input must be complemented with material tools and personal experience.

References

- A quote by Charlie Parker*. (n.d.). <https://www.goodreads.com/quotes/19049-music-is-your-own-experience-your-thoughts-your-wisdom-if>
- Lobo, I. (2008). *Epistasis: gene interaction and complex disease phenotypes* | *Learn Science at Scitable*. <https://www.nature.com/scitable/topicpage/epistasis-gene-interaction-and-the-phenotypic-expression-907/>
- Qizi, M. G. A., & Jumaeva, N. K. Q. (2024, December 4). *A Study of Language in the Preservation of Religious Texts*. <https://scientific-jl.org/mod/article/view/4757>
- Warmerdam, V. D. (n.d.). *High on probability, low on confidence - Koaning.io*. <https://koaning.io/posts/high-on-probability-low-on-certainty/>